

Highlights

Numerical-Decidability-Gated Metamorphic Testing for Scientific Machine Learning Surrogates

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- Numerical decidability gates surrogate metamorphic-test verdicts.
- Operator-floor bounds decide when divergence checks are measurable.
- Relation records, verdicts, and ledgers make admissibility auditable.
- Bounded executions show coverage follows the admitted relation set.
- Results support complementarity, not baseline superiority.

Numerical-Decidability-Gated Metamorphic Testing for Scientific Machine Learning Surrogates

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Abstract

Scientific machine-learning (SciML) surrogates are hard to test because exact outputs are unavailable for transformed inputs. Metamorphic testing can turn physical relations into oracle-free tests, but a candidate relation is not automatically an interpretable verdict: a failure may reflect a surrogate inconsistency, an invalid relation application, or a numerical artifact of the measurement operator. This paper treats numerical decidability as an admissibility condition for SciML metamorphic testing. We define a four-condition admissibility predicate covering physical basis, transformation preconditions, representation mapping, and whether the verdict tolerance dominates the operator’s intrinsic floor. The predicate is implemented through metamorphic-relation (MR) records, executable checks, typed verdicts, and a fail-closed ledger. We give a discrete-divergence floor bound and instantiate it on the cylinder-flow mesh, with a second topology used as bounded stability evidence. The gate changes verdicts rather than only annotating them: it defers absolute conservation when the floor dominates, rejects an inadmissible airfoil continuity relation on physical-basis grounds, and keeps node permutation as an implementation sanity check. Bounded MeshGraphNets, PointMLP, physics-informed neural-network (PINN), Fourier-neural-operator (FNO), airfoil, and cross-program executions show that detector coverage follows the admitted relation set. The evidence supports an au-

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ditable admissibility workflow, not general SciML reliability, baseline superiority, arbitrary-mesh soundness, or real-world defect-detection rates.

Keywords: Metamorphic testing, oracle problem, scientific machine learning, software testing, verification and validation

1. Introduction

Scientific machine-learning (SciML) surrogates are now used as software components for approximating expensive simulations [1, 2, 3]. Mesh-based neural simulators such as MeshGraphNets (MGN) predict physical fields on irregular meshes [4], but they also inherit a classical testing difficulty: for many candidate inputs there is no cheap exact output against which the program can be checked, the test oracle problem [5]. Held-out rollout error is therefore necessary, but it does not by itself tell a tester whether the software respects relations that should hold under physically meaningful transformations.

Metamorphic testing (MT) addresses the oracle problem by checking relations among multiple executions rather than a single expected output [6, 7]. Conservation laws, symmetry relations, nondimensional similarity, and continuity conditions can all suggest candidate metamorphic relations (MRs) for scientific software [8, 9, 10]. The testing problem considered in this paper begins after such candidates have been proposed. A physics-derived relation is not automatically a valid software test: mirror symmetry may depend on geometry and boundary labels; a divergence relation may depend on a discrete operator, mesh weights, and boundary treatment; a transformed case may leave the relation’s domain before it reveals any system-under-test (SUT) inconsistency.

For SciML surrogate software, this creates an oracle-free verification-and-validation (V&V) interpretability problem: when an MR check fails, the tester must ask whether the SUT violated a genuine relation, the relation was applied outside its validity conditions, the output mapping was incompatible with the software representation, or the scoring operator could not resolve the claimed effect. Treating all such outcomes as ordinary pass/fail MT verdicts makes the evidence hard to audit and easy to overclaim.

Here, numerical decidability is an admissibility condition for SciML MR use: before a candidate relation can serve as an oracle-free test, its verdict must be resolvable for the deployed discretization and measurement opera-

tor. The paper does not claim to invent MT, and it does not replace residual, uncertainty, accuracy, or trust-region diagnostics. It treats numerical decidability as a gate on relation admissibility, implements the gate through relation records, executable checks, and typed verdicts, and reports how the gate changes testing outcomes. A downgraded or rejected relation can still be reported as OOD stress evidence; it is just not licensed as a SUT fault verdict. Physical knowledge, expert reasoning, large language model (LLM)-assisted candidate lists, and the NOETHER two-layer MetaPattern model [11] may suggest candidate relations, but the evidence supports bounded SciML V&V utility rather than general SciML reliability or real-world defect-detection rates.

Two principles organize the method. First, an MR is *admissible* only when its validity conditions hold for the SUT and the verdict is numerically decidable: the tolerance must dominate the intrinsic floor of the measuring operator (machine precision, a mapping floor, or a discretization floor, by relation type). Second, relation-level verdicts are interpreted along two axes, relation-violation and domain-violation magnitude, separating SUT inconsistency from out-of-relation-domain application, numerical-floor deferral, and inconclusive evidence. We refer to the resulting framework as numerical-decidability-gated, relation-indexed SciML V&V. The main empirical setting is MeshGraphNets-family cylinder flow, with the same admissibility predicate additionally exercised on a second computational fluid dynamics (CFD) task and additional partial differential equation (PDE) families; these are bounded case studies, not cross-domain reliability estimates.

We do not claim a completed applicability map. The evidence is one bounded within-SUT contribution plus bounded witnesses; a calibrated cross-SUT map is left to future work.

Contributions. The single methodological idea is to gate a candidate relation’s admissibility on numerical decidability: a relation is admitted only when its verdict tolerance dominates the measuring operator’s own intrinsic error floor, a test that to our knowledge is new for learned SciML surrogates evaluated out of distribution (OOD). The contributions below make this admissibility condition explicit, executable, and inspectable; the research questions and evidence mapping are defined in the experimental design (Table 5). The positive claim is methodological: a numerical-decidability admissibility gate, a typed domain-inadmissibility verdict, and seeded-fault diagnostic stress tests whose MR-class-to-fault-class readings are reported as stress-test evidence rather than validated localization models.

Numerical-decidability criterion. The head contribution is an admissibility condition for SciML MR testing: a relation-level verdict is admissible only when its tolerance dominates the intrinsic numerical floor of the measurement operator. For the deployed cylinder-flow divergence check, this criterion is instantiated by a discrete-divergence floor bound, a measured floor matched by a closed-form predictor, a dominating a-priori upper bound, and a second-topology stability check (Sec. 5.5). The detailed statement is deliberately kept with the operator-floor experiment, where the operator class, mesh assumptions, and out-of-scope cases are defined before use.

Auditable implementation mechanism. The criterion is implemented as auditable executable checks. Each admitted relation is represented as a relation record and executable check that records the transformation, mapping, metric, tolerance, exclusion rule, and ledger fields needed for inspection. The record is not a separate methodological concept.

Typed relation-level verdicts. The verdict scheme is the criterion’s interpretation layer. It separates relation violation from domain violation, instantiating for physics-governed SciML the constraint-architecture pattern behind bug-versus-inapplicability reasoning [12, 13] and MetaTrimmer [14]. The added element is a typed classification of physical, geometric, boundary-condition, mapping, and operator-admissibility limits.

Bounded necessity evidence. The case studies evaluate what the criterion changes beyond rollout accuracy, generic MR generation, and LLM-assisted MR candidate elicitation, drawing primarily on MeshGraphNets cylinder-flow runs with supporting airfoil, physics-informed neural network (PINN), Fourier neural operator (FNO), and cross-program checks. The evidence is reported as complementarity and measured gate value, not baseline superiority (Section 5). The admitted MR set is also used to interpret fault-detection boundaries and blind spots, but this SciML-specific coverage reading is qualitative and bounded rather than a general fault-detection-rate claim.

Paper organization. Section 2 positions the work against oracle-free testing, physics-based MT, and SciML diagnostics. Section 3 defines the gate, relation record, executable check, and verdicts. Section 4 states the research questions, subjects, metrics, baselines, and analysis plan. Section 5 reports the empirical evidence and claim boundaries, followed by threats to validity, future work, and conclusions.

2. Related Work

2.1. Oracle-free testing and MR candidate sources

Mesh-based neural simulators learn dynamics on graph or mesh representations of physical systems. MeshGraphNets encodes mesh nodes and edges, propagates information by message passing, and predicts future physical fields through autoregressive rollout [4]. In this paper these models are treated as software SUTs. The testing question is whether a rollout is accurate on held-out trajectories and whether the SUT preserves declared relations under controlled transformations and explicit tolerances.

Metamorphic testing (MT) addresses programs lacking a per-input oracle [6, 7, 15]. Rather than checking each output against an expected value, MT checks a relation among source and follow-up executions. This framing has been applied to scientific software, PDE programs, simulations, and machine-learning (ML) systems [16, 17, 8, 10], including recent applications to tabular ML such as credit scoring [18]. For SciML surrogates, however, an MR is not merely an input perturbation: its validity depends on governing equations, boundary conditions, mesh representation, output mapping, and numerical tolerance. This makes MR admissibility part of the oracle construction problem itself.

Testing scientific software without a reliable oracle is a long-standing problem, and metamorphic testing is the most developed response in the surveyed literature [19]. MRs can be elicited from monotonicity, conservation, scaling, and domain-specific expectations [8, 9, 10, 20]. Predictive MR identification has been pursued with graph-kernel learning over scientific source code [21], by separating input and output patterns and by image-based output-pattern recognition for numerical solver codes [22, 23], and with large language models that discover metamorphic-relation variables for scientific software [24]. Symmetry-based MRs paired with statistical violation criteria have also been applied to deep trajectory predictors [25, 26]. Data-driven search can discover candidate transformations in complex physical software [27], and design assumptions can be encoded as MRs for cyber-physical-system (CPS) testing under control-plant invariants [28].

Separate work has made MR specification and applicability more explicit. General MT tooling has explored machine-readable MR representations, constraint definition, test-data-driven MR selection, and general MR specification languages [13, 29, 30]. These systems strengthen MR engineering in the large, but they do not by themselves decide whether a physics-derived

relation is numerically measurable, in-domain for a transformed CFD case, or interpretable as a SciML V&V verdict.

These approaches can supply candidate sources, but candidate identification is not admissible SciML test construction. MR identification, including our own prior physics-derived MR construction for numerical solver and nuclear-power software [31, 22], is established work; this paper takes candidate MRs as input and contributes the admissibility gate rather than a new identification method. Fault-oriented MR work ranks relations by observed detection power [32, 33]; the present paper uses the admitted relation set later to interpret bounded coverage and blind spots, but treats that as a consequence of admissibility rather than as the primary novelty.

2.2. *Physics-based MT, tolerance, and applicability*

Three contemporary strands are closest to this paper. Yang and Chui [34] and Reichert et al. [35] applied physics-derived MRs to hydrologic ML or conceptual models, showing that learned-model accuracy and relation consistency can diverge and that calibration or forcing regimes affect interpretation. Eniser et al. [36] introduced *relaxations*, numerical tolerances embedded in MR oracles, for stochastic reinforcement-learning (RL) policy testing. Duque-Torres et al. [12, 13, 29] and MetaTrimmer [14] address bug-versus-inapplicability reasoning with explicit MR constraints and automated constraint derivation.

The present work combines these ideas in a SciML-specific V&V workflow. From physics-based MR testing it takes relation-level checks; from relaxations it takes the need for calibrated tolerances; from violation-attribution work it takes the need to avoid treating inapplicability as a SUT failure. What none of these strands supplies is the central condition used here: admissibility made contingent on numerical decidability. Relaxations attach a tolerance to an oracle, and constraint- or attribution-based work decides whether a relation applies to an input, but neither requires the verdict tolerance to dominate the measuring operator’s intrinsic error floor, a property of the discretization fixed before any fault or SUT is considered. The floor gate, not the recording discipline, is what is genuinely new here. The admissibility decision, tolerance rationale, executable check, and typed verdict are then recorded together and tied to evidence artifacts. Table 1 positions the paper against the closest-prior capabilities.

Table 1: Closest-prior capability matrix.

Approach	What it supplies	What remains missing for the evaluation objective	What this paper adds
Scientific-software MR identification	Candidate relations from domain rules, source-code learning, LLM variables, and data-driven search	Candidate validity, numerical decidability, executable check format, and typed SciML verdicts	Treats these methods as candidate sources, not validity authorities
Reichert et al.	Physics-derived MRs and informal filtering in a learned hydrologic surrogate	Explicit admissibility predicate, operator-floor tolerance, reusable relation records, and typed verdicts	Formalizes the candidate-to-verdict path for mesh-based SciML surrogates
Eniser et al.	Relaxed MR oracles with numerical thresholds	Measurement-operator floor and relation-domain interpretation for deterministic SciML surrogates	Grounds tolerance in the scoring operator’s own numerical floor
Duque-Torres / MetaTrimmer	Binary bug-versus-inapplicability separation with explicit MR constraints and automated constraint derivation	A typed inadmissibility verdict and operator-floor admissibility for physics-governed SciML	Replaces the binary skip/proceed pre-filter with a two-axis typed verdict tied to physical, geometric, boundary, and operator limits
Formal MR specification and constraint tooling	Machine-readable MR descriptions, constraint phases, and supporting systems	Physics-specific admissibility dimensions, measurement floors, and executable evidence ledgers for SciML	Uses specification ideas but binds each verdict to relation-domain and numerical-decidability evidence
This paper	Physics, expert, and LLM-assisted candidates plus artifact-backed execution	Bounded case-study evidence, not general SciML reliability or baseline superiority	Constructs relation records, runners, ledgers, and typed verdicts under a four-condition gate

2.3. *SciML diagnostics and positioning*

SciML reliability research has developed complementary tools: physics residuals, uncertainty quantification (UQ), conformal prediction, certified error bounds, and failure-mode analysis [1, 2, 3, 37, 38]. PINN training can exhibit gradient-flow pathologies that bias trained models away from the governing residual [39, 37], sharpening the need for checks that do not assume a converged surrogate. These diagnostics are not competitors to MT. A residual, equivariance error, or uncertainty score becomes an executable MR only when paired with a source case, follow-up transformation, tolerance rule, exclusion rule, and verdict interpretation. The proposed workflow uses such diagnostics as possible measurements inside a multi-execution oracle-free check.

Hybrid ML-solver frameworks use residual thresholds or error estimates to decide when a learned component should be trusted [40, 41]. Such systems operationalize runtime trust regions, whereas this study acts offline, applying physically derived transformations before deployment to estimate where relation violations occur.

The distinction from residual- and uncertainty-based trust estimation is deliberate. UQ, conformal prediction, residual-threshold trust regions, and physical-consistency diagnostics that score a physically derived residual on observed outputs [42] locate unreliable behavior from observed inputs. The present method acts in relation space: it applies a controlled transformation and reports which necessary relation breaks under it, indexed by that transformation. The claim is complementarity, not superiority: relation-indexed evidence answers a different question from scalar accuracy, residual magnitude, or uncertainty.

The preceding subsections reviewed metamorphic testing, MR identification, scientific-software MT, residual and uncertainty diagnostics, LLM-assisted candidate generation, and the NOETHER two-layer MetaPattern model as sources of testing ideas. The contribution here is narrower: SciML MR use should be treated as an admissibility and executable-check problem. A candidate relation becomes a useful oracle-free V&V check only after its physical basis, transformation preconditions, output mapping, tolerance, exclusion rule, executable check, and relation-level verdict are recorded. Recent work derives or refines domain-specific MRs for scientific programs directly from their numerical models [43, 44]; the wedge here is the learned surrogate evaluated out of distribution and the requirement that a relation’s tolerance

dominate the measuring operator’s numerical floor, rather than testing a numerical solver against derived relations. The missing combination is thus an admissibility-gated, executable, and auditable workflow for physics-governed SciML surrogate software (Table 1).

3. Method

3.1. Overview

The method turns a physics-derived candidate relation into an auditable V&V check only after three steps are made explicit: admissibility, executable checking, and verdict interpretation. These three design roles are evaluated by the research questions introduced in Section 4.1. Figure 1 summarizes the workflow. Candidate generation is deliberately separated from validity judgment: expert reasoning, physical laws, the NOETHER two-layer MetaP-pattern model, and LLM-assisted lists may propose relations, but none certifies a relation as valid for a specific SUT.

Definitions. Let $r = (b, T, M, m, \tau, P)$ denote a candidate relation. Here b is the physical or software basis, T is the source-to-follow-up transformation, M maps follow-up outputs back into the comparison space, m is the metric, τ is the verdict tolerance, and P is the set of source and follow-up preconditions. For a SUT s and source case x , the admissibility gate returns

$$G(r, s, x) \in \{\text{admit, reject, stress, defer}\}.$$

If the relation is admitted, or retained only as stress evidence, the executable check returns $E(r, s, x) = (y, y', z)$, where y and y' are the source and follow-up outputs and z is the measured relation evidence. A verdict map $V(G, z)$ then assigns the typed verdict. Only an admitted relation with an in-domain fail verdict may support a SUT-inconsistency claim; rejected, stress-only, deferred, or inconclusive outcomes remain evidence about the relation, the domain, or the measurement setup.

Terminology and verdict types. A verdict is read on two axes, relation-violation and domain-violation magnitude (detailed in Sec. 3.5). A candidate relation is accordingly *admitted* (a large in-domain violation reads as SUT inconsistency), *rejected* on physical grounds, *deferred* when no tolerance dominates the operator floor, *out-of-relation-domain*, or *inconclusive*. Four MR families recur in the case studies: node-permutation equivariance, mirror-y reflection symmetry, discrete-divergence continuity/conservation,

and exact mirror-y on a constructed symmetric mesh. As a running example, a mirror-y candidate is first checked against geometry and boundary labels; on an asymmetric mesh it becomes OOD stress rather than a fault oracle, while on a constructed symmetric mesh it remains an admitted equivariance test.

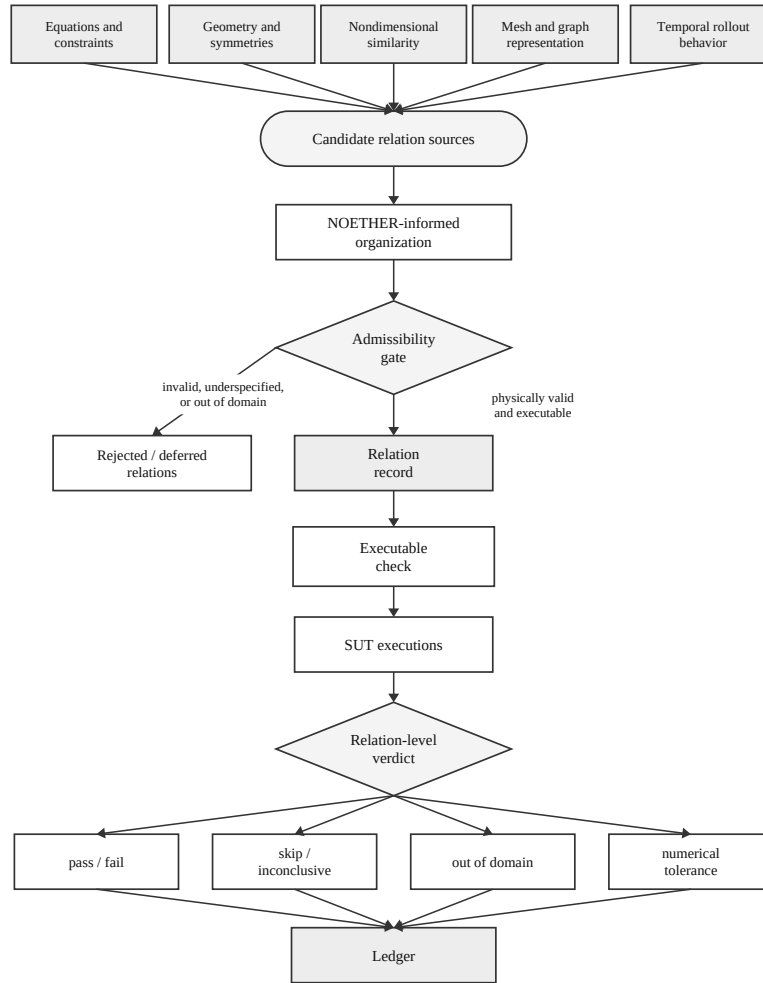


Figure 1: Admissibility-gated V&V workflow.

3.2. Candidate relation sources and organization

This paper applies two prior MR-identification results rather than redefining MR identification. The four-layer classification of scientific-computing MRs [45] asks where relations come from: physical, computational, code, or data layers. Its physical, computational, and code layers can yield real MRs when the corresponding semantic constraint is established. The two-layer MetaPattern model of NOETHER [11], a same-author preprint, asks a different question: what algebraic shape a candidate relation has. We use these two results to source and organize candidate relations, and use NOETHER only as an optional scaffold for generating and naming likely MR candidates. The central decision point in this paper is therefore not candidate discovery itself but the admissibility gate of Section 3.3, which decides whether any candidate becomes an admissible MR for a specific SUT.

Layer 1 (MetaPattern). A *MetaPattern* is the equivalence class of MRs generated, through TRANSLATE, from one minimal algebraic structure of the program-induced operator algebra $\mathcal{A}_P$. The five MetaPatterns are generated by a group action (G), partial order ($O_{\leq}$), self-adjoint operator (T^*), time-reversal involution ($\mathcal{T}_{\text{rev}}^*$), or parametrized limit ($\mathcal{L}^*$); their set is $\mathbb{M}(\mathcal{A}_P)$. Conservation is therefore the group-action member m_{inv} .

Layer 2 (MR family). An *MR family* instantiates one MetaPattern under a fixed transformation template and mode. Families use the form $f_{\text{parent.child}}$; legacy letters a–j remain cross-reference anchors. The map is one-to-many: $G \rightarrow \{f_{\text{inv.eqv}}, f_{\text{inv.con}}\}$, $T^* \rightarrow \{f_{\text{adj.self}}, f_{\text{adj.dual}}\}$, $\mathcal{T}_{\text{rev}}^* \rightarrow \{f_{\text{rev.traj}}\}$, $O_{\leq} \rightarrow \{f_{\text{mono.stat}}, f_{\text{mono.shape}}\}$, and $\mathcal{L}^* \rightarrow \{f_{\text{conv.lim}}, f_{\text{conv.rate}}, f_{\text{conv.repr}}\}$.

For the cylinder-flow surrogates studied here, the instantiated families are $f_{\text{inv.eqv}}$ (equivariance, a), mirror-y reflection; $f_{\text{inv.con}}$ (conservation, b), discrete-divergence continuity and, on the airfoil task, compressible mass balance; $f_{\text{conv.repr}}$ (representation-invariance, j), node-permutation and encoding contracts; and $f_{\text{conv.lim}}$ and $f_{\text{conv.rate}}$ (convergence, accuracy-order, h–i), the operator-floor mesh-refinement relation. The $f_{\text{mono.stat}}$ (static-order, f) Reynolds–Strouhal monotonicity is a candidate not exercised here, $f_{\text{rev.traj}}$ (trajectory-reversal, e) is empty by construction for viscous dissipative flow, and the self-adjoint families $f_{\text{adj.self}}$ and $f_{\text{adj.dual}}$ (c–d) appear only in the cross-family PINN and FNO extensions. The same relation may be admissible as one family yet inadmissible once instantiated for a given SUT: mirror symmetry holds on a symmetric domain but not on an asymmetric mesh with incompatible boundary labels, and conservation is a physical expectation

but not an executable verdict when the discrete measuring operator has an unresolved floor. The admissibility gate records where each relation survives.

3.3. *Admissibility gate*

A candidate relation is an *admissible MR* only when four conditions hold together. First, it must have a physical or software basis. Second, its transformation preconditions must hold for the source and follow-up cases. Third, boundary conditions and output mappings must remain compatible after transformation. Fourth, the verdict must be numerically decidable: the tolerance must dominate the intrinsic error floor of the measuring operator. None of these conditions is optional. Table 2 records the gate, the relation-record fields, and the verdict semantics.

Table 2: Admissibility gate and verdict semantics.

Method element	Required record	Allowed interpretation	Design role
Basis	Equation, boundary condition, nondimensional law, representation property, numerical assumption, or implementation contract	Relation may be considered only for SUTs where this basis applies	Gate
Preconditions	Source-case constraints, follow-up transformation, preserved quantities, and exclusion rule	Missing preconditions produce skip or out-of-relation-domain, not SUT failure	Gate
Mapping compatibility	Boundary-label compatibility and output mapping from follow-up output back to source coordinates/fields	Incompatible mapping rejects or downgrades the candidate	Gate
Numerical decidability	Metric, tolerance, tolerance provenance, and measurement-floor evidence	A verdict is meaningful only when tolerance dominates the operator floor	Gate
Relation record	Identifier, source category, transformation, mapping, metric, tolerance, exclusions, expected verdicts, and schema	The admitted relation becomes a reusable executable check	Check
Ledger	Source/follow-up run ids, metric outputs, floor evidence, exclusions, logs, and verdict	The reported claim is auditable from evidence rather than prose alone	Ledger
Verdict semantics	Pass, fail, skip, out-of-relation-domain, numerical-tolerance issue, or inconclusive	Only fail inside the valid domain may be read as SUT inconsistency	Verdict

The gate produces four design-time decisions: admit as an executable MR; downgrade to OOD-stress when the relation is physically informative but outside its exact domain; reject when basis, preconditions, or mapping fail; or defer when missing numerical evidence prevents a verdict. For example, node permutation can be admitted as a representation MR when the adapter and inverse mapping are deterministic; mirror-y is conditional on compatible geometry, boundary labels, and vector-component mapping; absolute discrete

divergence is deferred when the reference field and measuring operator floor cannot support an absolute verdict.

3.4. Relation record and executable check format

A retained MR is represented as a relation record containing the fields listed in Table 2 plus interpretation limits. The record derives from the five core concepts; the executable check operationalizes it.

Each executable check includes transformation code, runner configuration, metric computation, verdict logic, and evidence recording. The ledger is part of the method rather than only replication packaging: every relation-level claim must be traceable to the source run, follow-up run, mapping, metric output, floor evidence, exclusion decision, and verdict.

3.5. Relation-level verdicts

An MR execution can produce: **pass** (relation holds within tolerance); **fail** (violated within the valid relation domain); **skip** (required precondition missing); **out-of-relation-domain** (the transformed case is outside the relation domain); **numerical-tolerance issue** (the measured effect is dominated by threshold or resolution uncertainty); or **inconclusive** (insufficient artifact).

These verdicts form the two-dimensional space in Figure 2. One axis is relation-violation magnitude, typically expressed as a violation-to-floor or violation-to-tolerance ratio. The other is domain-violation magnitude: how far the transformed case lies outside the relation’s validity domain. Low domain violation with high relation violation is the region that may be read as SUT inconsistency; high domain violation is out-of-relation-domain; a violation inside the error floor is a numerical-tolerance issue. This decomposition defines the method’s verdict-interpretation role, which is evaluated explicitly in Section 4.1.

In this study, the domain-violation coordinate is operationalized per relation from committed measurements. For mirror-y, $D = m/(m + 1)$, where m is the worst reflected-node placement error in median-edge-length units. The synthetic symmetric mesh scores $D \approx 0$, whereas the real asymmetric evaluation mesh scores $D = 0.51$. Other MR families use their own precondition measurements. These D values denote structural placement within each relation’s own validity domain: D is a per-relation normalized coordinate, not a cross-relation calibrated metric, and the values cannot be averaged or ranked

across MR families; read per relation, not as a calibrated boundary measurement. A cross-relation calibration would need to normalize each family by its own admissibility boundary, mapping every gate threshold to a common $D_{\text{cal}} = 1$; that calibration is future work.

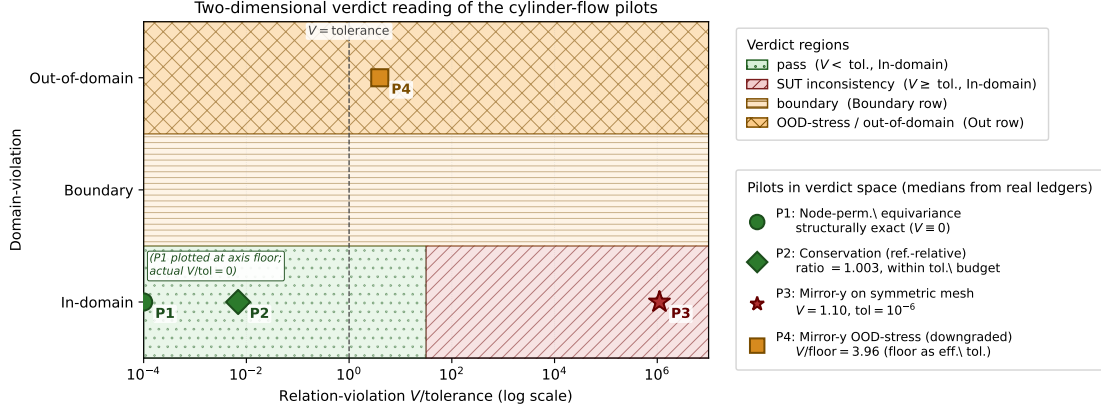


Figure 2: Two-dimensional relation and domain verdict space.

Retained MRs are interpreted through their two-layer position (Sec. 3.2): a relation’s MetaPattern and MR family fix which algebraic invariant it measures, and therefore which faults it can and cannot surface. This is an interpretation protocol, not a validated localization model; the seeded-fault results later are a bounded stress test of it.

Practitioner checklist and adoption boundary. For a new SUT: (1) state the relation and source; (2) record source/follow-up preconditions; (3) check boundary labels and output mapping; (4) bound or calibrate the operator floor; (5) compare verdict tolerance with that floor; and (6) issue a typed verdict and ledger the run. The method does not remove domain expertise or automate MR discovery. Adoption costs are relation-specific physics review, mapping code, floor evidence, and bookkeeping; when any item is missing, the outcome is reject, defer, or OOD-stress, without a fault claim.

4. Experimental Design

4.1. Research questions

The experimental design is organized around the following objective: to evaluate whether physics-derived MRs for SciML surrogate software can be

transformed into auditable, executable, and interpretable oracle-free V&V checks that distinguish genuine SUT inconsistency from relation invalidity and numerical measurement limits.

This objective is decomposed into four research questions:

- RQ1 – Admissibility.** What validity conditions determine whether a physics-derived candidate MR is admissible as an oracle-free test for a specific SciML surrogate SUT?
- RQ2 – Records and checks.** How can admissible MRs be represented as reusable relation records and executable checks with explicit transformations, mappings, metrics, tolerances, exclusions, and evidence records?
- RQ3 – Verdict interpretation.** How can relation-level verdicts separate SUT inconsistency from out-of-relation-domain application, numerical tolerance limits, and inconclusive evidence?
- RQ4 – Empirical utility.** In bounded SciML surrogate case studies, what evidence does the admissibility-gated workflow provide beyond rollout accuracy, generic MR generation, and LLM-assisted candidate elicitation?

4.2. Experimental subjects

The subjects are selected for evidence roles rather than for statistical representativeness. A subject is included only when it contributes at least one design need: a primary end-to-end SciML surrogate setting with executable checks; a changed physical task or architecture that can falsify a checklist-only reading of the gate; a cross-family PDE/neural-operator setting that exercises the same admissibility predicate; or a bounded defect/coverage witness that ties MR responses to committed fault or true-fault-class evidence. The study does not select subjects by repository popularity, lines of code, or issue/changelog-mined defects; those criteria would require a separate real-defect study. The evidence below therefore supports bounded V&V utility and coverage interpretation, not population-wide defect-detection effectiveness. Here, K is roster size.

In aggregate, the executable subject inventory comprises one primary CFD task, one second CFD task, same-task MGN variants, one PointMLP SUT, PhysicsNeMo executions, PINN and FNO rosters over two PDEs,

one independent periodic-advection primary workflow, and five external central-processing-unit (CPU)-only SUT replays. The fault-evidence inventory comprises 10 canonical MGN mutants, two adversarial MGN mutants, 48 PINN/FNO output-level probes, a PointMLP 20-fault catalogue, and the external sibling’s 70 witness rows and 20 true-fault-class rows. These counts are denominators for evidence scope; they are not sampling frames for a general defect-detection rate.

Table 3: Subject selection and evidence scope.

Evidence block	Subjects and evidence role	Boundary
Cylinder-flow mesh surrogates	Primary executable workflow plus same-task implementation checks: MGN K=6 roster over three trajectories, S4/S5 variants, PointMLP, and PhysicsNeMo MeshGraphNet ; seeded-fault stress uses 10 canonical MGN mutants, two adversarial mutants, and PointMLP 20-fault catalogue	Same task and bounded rosters; no real-defect rate or LOC-based representativeness
Airfoil mesh surrogate	Changed-physics gate discrimination on official DeepMind airfoil with PhysicsNeMo MeshGraphNet , K=6 checkpoints over 10 test trajectories	Typed inadmissibility evidence, not state-of-the-art airfoil accuracy or official-checkpoint evidence
PINN/FNO and periodic PDE workflows	Cross-family PINN/FNO Burgers/heat predicate checks plus independent periodic-advection primary workflow with six periodic-convolution SUTs and 60 relation cells per admitted MR	Synthetic/generated PDE evidence; not production CFD, real-defect, reliability, or broad neural-operator claims
External sibling programs	Minimum-MR-SubSet audit across seven program types plus five CPU-only end-to-end replays, including OpenMC and three SciML/PDE PASS_WITNESS rows	Read-only/sibling provenance; no new issue/changelog mining in this paper

Primary evidence comes from two bounded full-chain workflows. The first is a MeshGraphNets-family cylinder-flow case study: a trained checkpoint, its K=6 roster, S4/S5 variants, a PointMLP coordinate network, and NVIDIA PhysicsNeMo **MeshGraphNet** on official DeepMind cylinder_flow TensorFlow Record (TFRecord) files. The second is an independent synthetic periodic-advection workflow: six NumPy-trained periodic-convolution SUTs, ten held-out fields per SUT, two admitted MRs, and one rejected candidate. In both workflows, the primary unit is a relation-level cell: relation record, SUT/checkpoint, source case, transformed follow-up case, metric, and verdict.

Supporting evidence checks whether the same admissibility predicate applies outside the primary mesh-surrogate setting: airfoil gate discrimination, a fifteen-seed PINN roster (K=15), and trained FNO-2D checkpoints over

Table 4: Independent evidence units: Nominal-N, effective-N, and inference permissions.

Evidence unit	SUT/task and MR/operator	Independence source	Evidence role	Inference allowed	Inference forbidden
Cylinder primary workflow	MGN cylinder-flow MRs	K=6 checkpoints, three trajectories, exact-symmetric inputs	Primary method execution	within-family typed verdict behavior	cross-SUT or population rates
Same-task architecture check	S4/S5, PointMLP, PhysicsNeMo cylinder	different impls. on one task	Implementation check	rules out single-codebase artifact	broad production-CFD or reliability claims
Changed-physics task	PhysicsNeMo airfoil gates	second CFD task, non-zero angle of attack	Gate discrimination	task-specific typed verdict change	state-of-the-art or official-checkpoint claims
PINN/FNO PDE families	Burgers/heat operators	different representation families and calibratable floors	Cross-family predicate check	executable verdicts where floors are calibratable	neural-operator or real-defect rates
Periodic-advection primary workflow	six periodic-convolution SUTs	independent synthetic PDE/task, K=6 and 10 fields/SUT	Independent full-chain check	admit/reject gate-to-verdict execution	production CFD, real-defect, reliability, or broad generalization
Fault and external witnesses	seeded/probe catalogues plus sibling rows	stress catalogues and read-only external artifacts	Blind-spot stress and external audit	admitted-set coverage geometry	real-world defect rate or superiority

2D Burgers and heat data.

Secondary evidence includes LLM-simulated expert MR design, generic MR-generation scope contrasts, LLM-assisted candidate generation, rollout-accuracy diagnostics, and external witness reruns from the read-only Minimum-MR-SubSet repository.

Stress-test evidence uses seeded-fault and cross-program checks to interpret where the admitted MR set can and cannot observe faults.

4.3. Evaluation metrics

Primary metrics are admissibility decision, executable MR rate, verdict distribution, violation magnitude relative to tolerance or floor, exclusion rate, and relation-level interpretation output. Secondary metrics include inter-rater agreement for candidate screening, flakiness rate, complementarity with rollout accuracy, and stress-test detection by seeded-fault class. These metrics are interpreted at the relation-cell level rather than as general reliability or defect-detection rates.

4.4. Experimental method

The experiments instantiate the candidate-to-check workflow of Section 3. The relations evaluated are the cylinder-flow MR families of Section 3.2: equivariance ($f_{\text{inv.eqv}}$) and conservation ($f_{\text{inv.con}}$) under m_{inv} , representation-invariance ($f_{\text{conv.repr}}$) and the convergence/accuracy-order pair ($f_{\text{conv.lim}}$, $f_{\text{conv.rate}}$) under m_{conv} ; the static-order ($f_{\text{mono.stat}}$) and self-adjoint ($f_{\text{adj.self}}$, $f_{\text{adj.dual}}$) families are exercised only as candidate or cross-family checks, and trajectory-reversal ($f_{\text{rev.traj}}$) is empty by construction. For each retained relation, the workflow records a relation record, source and follow-up executions, transformation and mapping artifacts, metric outputs, floor evidence, exclusions, and a typed verdict. The design separates four evidence tiers: primary, supporting, secondary, and stress-test evidence.

4.5. Comparison baselines

Rollout accuracy, residual/UQ diagnostics, expert-designed candidates, generic MR generators, and LLM-assisted candidates contextualize RQ4. Rollout accuracy asks whether a prediction is close to observed held-out data; an MR asks whether a declared relation holds under a controlled transformation. The baseline contrast is therefore complementarity and gate value: false-alarm removal, typed triage, auditability, and executable-check reuse, not superiority in a shared scalar score. LLM and generic MR outputs are treated as candidate sources; the admissibility gate, not the generator, decides whether a candidate becomes an executable MR.

4.6. Analysis plan and evidence mapping

Table 5 maps research questions to evaluation evidence, artifacts, units of analysis, and claim boundaries; it is a design rather than a result table.

Table 5: Evaluation design.

Item	Evaluation question	Evidence tier and unit	Artifacts	Claim boundary
Goal	Does the full workflow connect candidate, gate, executable check, verdict, and evidence?	Primary workflow trace; relation-level cell	Relation records, runners, claim ledger	Record/check construction only; no general reliability claim
RQ1	Which candidates are admissible, rejected, downgraded, or deferred?	Primary and supporting gates; candidate/MR	Gate records, operator-floor sweep, exclusions	Physical validity is relation- and SUT-specific
RQ2	Are admitted relations executable and auditable?	Primary execution cells; source/follow-up pair	Transformations, runner configs, metric files, logs	Execution completeness, not complete automation of MR discovery
RQ3	Do verdicts separate SUT inconsistency from domain and numerical limits?	Verdict ledger; MR-by-case cell	Two-axis verdicts, metric/floor ratios, exclusion logs	Per-relation interpretation; no cross-relation calibrated domain score
RQ4	What evidence is added beyond rollout accuracy and candidate-generation comparators?	Primary, supporting, secondary, and stress-test tiers	Rollout diagnostics, LLM/generic baselines, witness reruns, seeded-fault ledgers	Complementarity and bounded utility; no baseline superiority
Impl.	What blind spots follow from the admitted MR set?	Stress-test tier; mutant/MR cell	Seeded-fault catalogue and cross-SUT removal check	Bounded coverage interpretation, not real-world detection rate

The study reports inferential statistics only where the sampling structure supports them, labeling descriptive counts otherwise. Binary detector outcomes over the unified fault catalogue carry Wilson 95% confidence intervals (CIs) per detector; continuous violation magnitudes use exact Wilcoxon signed-rank tests with a rank-based effect size (Cliff’s δ), where the design supports paired comparison; roster medians carry paired bootstrap intervals; and the operator-floor sweep reports a log-log slope with a 95% CI. Repeated cells from one trajectory or one checkpoint family are reported as descriptive cell summaries, not independent population samples.

4.7. Ethics, integrity, and reproducibility

The study does not involve human subjects or personal data. SUT versions, datasets, checkpoints, scripts, and run logs are recorded when licensing permits. Failed, skipped, out-of-relation-domain, numerical-tolerance, and inconclusive outcomes remain in the ledger. LLM use is restricted to candidate generation and evidence organization, not validity judgment. No candidate MR is described as valid without a passing gate decision and relation

record; no OOD violation is described as a program fault without supporting verdict evidence.

5. Results and Discussion

Evidence is reported by research question rather than execution chronology, following Table 5. Evidence roles keep breadth from becoming representativeness: the cylinder-flow and periodic-advection workflows are the two bounded primary full-chain executions; airfoil, PINN, and FNO are supporting falsification checks of whether the same gate changes verdicts under changed physics or operators; LLM/generic baselines and sibling reruns are secondary scope contrasts; and seeded faults are stress tests of detector blind spots. Baseline contrasts in Appendix Appendix C measure the gate’s value, not method superiority: gate-admitted detectors stay clean on the correct SUT where gate-rejected templates false-alarm, and rollout accuracy is complementary rather than defeated. The tables likewise have distinct jobs: Table 5 gives the design, Table 6 gives relation-level verdicts, Table 7 summarizes primary results, and Table 4 limits inference.

5.1. Evidence overview

The compact claim-to-evidence map binds every claim to a tracked artifact using reviewer-facing claim names; the full table is in Appendix A (Table A1), and the authoritative runtime mapping remains in `claim-ledger.yml`.

Table 6 is a compact verdict index; detailed claim paths and per-run artifacts are in Appendix Appendix A and `claim-ledger.yml`.

Table 6: Relation-record verdict map.

Evidence block	Gate outcome	Runtime verdict	Boundary
Cylinder-flow MGN/variants	Node permutation admitted; mirror exact relation downgraded or admitted only on symmetric mesh; absolute conservation deferred	Node permutation exact; mirror OOD stress fails 180/180; exact symmetric-mesh mirror fails 18/18; reference-relative conservation passes 162/162	Primary bounded workflow; absolute conservation remains deferred; no geometry-independent or population rate
Same-task architecture checks	Same predicate applied to S4/S5, PointMLP, and PhysicsNeMo cylinder executions	S4/S5 and PointMLP reproduce typed verdict pattern; PhysicsNeMo smoke records node-permutation pass and mirror OOD stress	Implementation artifact check, not production-scale reliability

Airfoil changed-physics task	Node permutation admitted; incompressible continuity and mirror-y rejected/deferred for physical/numerical reasons	60/60 node-permutation exact; continuity and mirror verdict types change with compressible/non-zero-angle physics	Gate discrimination, not state-of-the-art airfoil accuracy
FNO primary workflow upgrade	Translation admitted; periodic discrete-conservation admitted-with-reference- floor; Dirichlet translation rejected	24/24 translation passes, 24/24 conservation failures, and 6/6 rejected Dirichlet exact-MR executions	Full gate-to-verdict FNO evidence, outside cylinder-flow and broad neural-operator claims
Periodic-advection primary workflow	Periodic translation and mass/mean conservation admitted; fixed-velocity mirror candidate rejected	60/60 translation passes, 60/60 mass-conservation passes, and 6/6 fixed-velocity mirror rejections	Independent synthetic PDE workflow; not production CFD or real-defect evidence

5.2. Primary cylinder-flow evidence

All pilots run on one real trained MeshGraphNets cylinder-flow surrogate, using the read-only Minimum-MR-SubSet checkpoint recorded in the experiment ledger. They exercise three different gate outcomes; raw outputs, manifests, and metric ledgers are committed (Table A1).

Table 7 gives a compact reading guide for the detailed evidence that follows. It is not a separate analysis; each row points to a relation-level result, its licensed interpretation, and the boundary that prevents overclaiming.

Representation MR. Node-permutation equivariance holds to machine precision (relative L_2 norm = 0.0 at tolerance 1e-6). This is a structural property of message-passing and is reported as a pipeline-implementation sanity check rather than model capability or accuracy evidence.

Geometric MR. The admissibility gate classifies exact mirror-y equivariance as out-of-relation-domain for this mesh (the reflection is non-bijective, the worst reflected-node mismatch is about one edge length, and the cylinder is off-centre by 7.2 mm) and downgrades it to an approximate nearest-neighbour OOD-stress probe scored against a same-space mapping-error floor. The probe failed on 10 of 10 recorded eval frames (median relative L_2 norm 0.737, median V/floor 3.96, floor range 3.02–5.55x), with no frame skipped or inconclusive. This is a bounded within-SUT frame-level OOD-stress result: one SUT, one checkpoint, one MR, one eval trajectory. The frames are consecutive, not independent samples, so the exact binomial 95% interval for 10/10 is wide ([0.69, 1.00]). The mapping-error floor derives from the same geometric mismatch that triggered the downgrade, so V/floor alone cannot separate a model-level violation from an amplified geometric artifact on this mesh. The symmetric-mesh run below resolves that ambiguity.

Table 7: Primary result summary.

Check	Observed result	Interpretation	Boundary
Node permutation	Relative L_2 norm = 0.0	Execution and inverse-mapping sanity check for a representation MR	Does not establish model accuracy or reliability
Asymmetric mirror-y	10/10 OOD-stress failures; median relative L_2 norm 0.737	The gate downgrades exact symmetry to stress evidence on this mesh	Not an exact symmetry verdict or geometry-independent fault claim
Continuity	Absolute conservation deferred; reference-relative ratio about 0.4–0.8%	Numerical floor blocks an absolute verdict while allowing a non-regression diagnostic	Not an absolute mass-conservation claim
Rollout contrast	One-step median relative L_2 norm 0.0216; mirror-y violation about 34x larger	MR evidence answers a relation question distinct from accuracy	The quantities are different relative- L_2 objects
Symmetric mesh	Exact mirror-y admitted and fails; relative L_2 norm 1.10	The out-of-sample gate admits the relation when geometry is bijective	Synthetic no-obstacle OOD geometry
Seeded faults	5/10 mutants detected; detections separate by MR family	Admitted MRs expose some invariant-breaking faults and blind spots	Stress-test evidence, not a real-world defect-detection rate

Continuity MR. A piecewise-linear (P1) discrete-divergence operator produces a non-negligible divergence even for the ground-truth field on this coarse mesh, so an absolute divergence-free tolerance is not calibratable and the absolute mass-conservation relation stays deferred. As a reference-relative diagnostic, the surrogate’s predicted next-state divergence stays within about 0.4–0.8% of the reference on the recorded eval frames; the interior-only ratio rules out a pure boundary-imposition artifact. This asserts no absolute conservation. Two caveats bound the diagnostic. First, the reference divergence is not yet decomposed into discrete-operator error, solver projection artifact, or genuinely non-solenoidal training data, so if the reference field is itself materially non-solenoidal the verdict is *inconclusive: reference-relative non-regression guard* rather than a conservation measurement. Second, the diagnostic uses a 50% regression threshold (ratio > 1.5) on two eval frames only, so “passes” means “does not regress conservation by more than 50% on those frames,” not “conserves mass.”

Rollout-accuracy diagnostic. On the same eval trajectory, the surrogate’s one-step next-state prediction error ($v_{\text{pred}} = v_t +$ denormalized predicted delta, the trainer’s own convention) has median relative L_2 norm 0.0216 (mean 0.044; min 0.0116, max 0.0788) over the nine recorded transi-

tions. The per-transition error is bimodal, so we quote both statistics. The mirror-y OOD-stress violation (median 0.737) is about 34 times the median accuracy and 17x the mean. The quantities are dimensionless relative L_2 -norm values of different objects, so this is an order-of-magnitude gap rather than a precise factor; it is one-step, not a free-running rollout-stability result.

Exact mirror-y on a symmetric mesh. We built a provably symmetric synthetic structured channel mesh whose reflection is a verified bijection (node-type match 1.0, reflection offset $< 10^{-12}$, edge set invariant), so the predicate admits exact mirror-y. On one constructed input the surrogate fails this exact equivariance test (relative L_2 norm 1.10). The normalizer control changes the violation only from 1.1032 to 1.1014, so the violation is dominated by learned message-passing weights. The boundary is synthetic, no-obstacle OOD geometry: the result confirms missing exact equivariance but is not a calibrated in-distribution magnitude.

Seeded-fault detection. We re-implemented, from the read-only Minimum-MR-SubSet witness taxonomy, an injected-fault catalogue of 10 pipeline faults across five fault classes (boundary-condition, mesh-adjacency, normalization-scale, temporal-rollout, physical-channel), and used the paper’s own MRs as detectors on the model’s predicted update. The continuity MR detected the two boundary-condition faults and the gross normalization fault (divergence ratio 3.8–10.6 vs the 1.5 threshold); the symmetry MR detected a physical-channel and a mesh-adjacency fault (violation rising 69–142% above baseline); node-permutation equivariance detected none, because these faults preserve node-relabeling invariance. 5 of 10 mutants were detected by at least one MR, and the detections separate by MR family (continuity to boundary/scale faults, symmetry to physical-channel/adjacency faults), a first bounded test of the interpretation protocol of Section 3.5, suggestive rather than a validation. Three boundaries: the detected faults are gross corruptions that any divergence- or symmetry-sensitive detector would catch; the edge-drop fault is a near-miss (32% mirror-y change vs the 50% threshold) and the boundary faults are invisible to mirror-y by construction; and the remaining undetected faults shift the absolute output without crossing the scored-quantity thresholds, delimiting where these MRs are structurally insensitive. Against a rollout-accuracy monitor on the same faults the two are complementary: the edge-permutation and channel-swap faults the symmetry MR catches leave the rollout error within $1.3\times$ of its in-distribution baseline (0.0216, under a $2\times$ regression threshold), so the MRs surface relation violations an accuracy monitor leaves within band,

while the gross boundary/normalization faults trip both and no fault is caught by accuracy alone (claim C38). It is one SUT, one checkpoint, one injected-fault catalogue.

The remaining subsections carry the same evidence at larger scope: the MGN roster tests denominator robustness, the operator-floor sweep calibrates numerical decidability, and the fault catalogue bounds detector geometry. Whether the gate transfers beyond the original MeshGraphNets workflow is tested by the cross-family PINN and FNO executions reported in Appendix Appendix C.

5.3. *Same-task replication*

The single-checkpoint evidence is replayed on a K=6 MeshGraphNets roster and three official DeepMind held-out test trajectories from the Minimum-MR-SubSet loader. S0 reuses the pilot checkpoint, so this is a replication/variant check rather than six independent SUTs. Node permutation is exact on all checkpoints. The primary empirical scope upgrade gives a K=6 x 3 trajectories x 10 mirror-y OOD-stress grid with 180/180 failures, a K=6 x 3 trajectories x 9 conservation-transition grid with 162/162 reference-relative passes while absolute conservation remains deferred, and a K=6 x 3 exact-symmetric-mesh input grid with 18/18 failures. The upgrade removes the single-source-trajectory denominator while remaining within one architecture family and dataset, so per-cell counts are descriptive summaries, not independent-trial inference, with the K=6 checkpoint roster as the effective independent unit.

Same-task multi-architecture replication. Three further cylinder-flow executions test whether the verdict pattern is an artifact of one implementation. A same-domain S4/S5 wider/deeper MeshGraphNet variant package reproduces the pattern (2/2 node-permutation passes, 60/60 mirror OOD failures, 54/54 conservation-diagnostic passes, 6/6 exact-symmetry failures). A newly trained PointMLP coordinate network (a different architecture class with no message passing) reproduces it as well (9/9 node-permutation passes, 10/10 mirror OOD failures, 9/9 conservation-diagnostic passes, 3/3 exact-symmetry failures, median rollout relative L_2 norm 0.0298). Third, the NVIDIA PhysicsNeMo MeshGraphNet production implementation, trained and evaluated on official DeepMind cylinder_flow TFRecords, reproduces the node-permutation pass and the mirror-y OOD-stress failure within a production framework rather than a research codebase. The same admissibility predicate, applied unchanged, produced the same typed decisions on

all three; this replicates the verdict pattern across architectures on one task and dataset, asserting no cross-dataset rate.

5.4. Second-task gate discrimination: compressible airfoil flow

A direct test of an admissibility gate is whether it produces the correct *different* verdict when the physics changes. We apply the identical four-condition predicate to a second CFD task on a second official dataset: the DeepMind **airfoil** benchmark (compressible transonic flow simulated with the SU2 solver, 5,233-node meshes with density, pressure, and velocity), distinct from incompressible cylinder flow. The PhysicsNeMo **MeshGraphNet** architecture has hidden width 128, 15 processor steps, and 2.33M parameters. It is trained in a bounded CPU/Apple Metal Performance Shaders (MPS)-scale workflow on the first 6 official train trajectories and evaluated across a K=6 checkpoint roster on the first 10 official test trajectories, producing 60 checkpoint-by-trajectory cells. The predicate produces a *different typed inadmissibility structure*. **Node-permutation equivariance is admitted and exact** (60/60, relative L_2 norm 0.0, training-independent representation contract). **Incompressible divergence-free continuity is rejected at the physical-basis gate** (condition i) because the density varies by a median factor of 2.25x across the field, so $\nabla \cdot u = 0$ is physically false here. On cylinder flow, the same relation passes conditions (i)–(iii) and is deferred only at condition (iv); the gate therefore assigns a categorically different verdict *type* for the physically correct reason. **Compressible unsteady mass conservation** $\partial_t \rho + \nabla \cdot (\rho u) = 0$ is the physically-correct replacement, its absolute discrete form deferred (uncalibratable operator floor) and a reference-relative diagnostic recorded (median residual ratio 1.19 across the K=6 roster). **Mirror-y chord symmetry is rejected at the boundary-precondition gate** (condition iii) because the SU2 trajectories carry a non-zero angle of attack. The one-step rollout error remains large at this bounded training budget and is reported only as a training-state diagnostic. These typed verdicts do not depend on rollout quality: admission rests on a representation contract, and rejection or deferral rests on physical and numerical reasoning, not surrogate outputs. This is a bounded second-task gate-discrimination result on the official MeshGraphNet architecture (claim C31), not an official-checkpoint, full-scale production, or state-of-the-art airfoil result.

5.5. Operator-floor resolution sweep

Proposition (C53: P1 operator-floor soundness). For P1 constant-per-cell divergence on shape-regular triangular meshes and C^2 divergence-free fields with Hessian bound M , the local floor is bounded by $C_{\text{shape}} M h$. An absolute divergence-free MR verdict is admissible only when its tolerance dominates this floor. Limited to this operator class: it gives no arbitrary/degenerate-mesh, non-P1, learned-output, boundary-mismatch, reliability, or fault-detection guarantee.

The P1 discrete-divergence sweep on symmetric meshes, over nine resolutions, gives slope 0.984 with 95% CI [0.975, 0.992] and $R^2 = 0.9999$, matching $O(h)$ (Figure 3). The general reason is local: for a shape-regular triangular mesh and a C^2 divergence-free reference field, P1 exactness on affine fields leaves only the second-order Taylor remainder, giving an operator-floor bound proportional to the Hessian scale and cell diameter (claim C53). The *absolute* floor is also closed-form for the concrete deployed-scale mesh: since the operator is exact on affine fields and the reference field is divergence-free, the measured floor equals the geometry-weighted second-order Taylor remainder exactly. A leading-order predictor from the analytic Hessian matches the measured floor to within 0.5% (1.337 vs 1.343), and a rigorous a-priori upper bound from the Hessian’s global spectral norm dominates it at every resolution. The same floor reproduces on a genuinely different topology — an unstructured Delaunay triangulation of a jittered grid (worst triangle angle about 14 degrees) over the same domain and reference field — with log-log slope 0.983 (Student-t 95% CI [0.953, 1.014], $R^2 = 0.999$), a matched-resolution unstructured-over-structured floor ratio of 1.06 median and 1.33 maximum, so the numerical-decidability verdict does not flip between the two topologies. The gate thus rests on a derived floor, not an empirical estimate, for this mesh; the floor is empirically topology-stable across these two mesh families, while arbitrary degenerate meshes, non-P1 operators, boundary-condition mismatch, and discontinuous fields remain outside the theorem. This is why absolute conservation remains deferred while the reference-relative non-regression guard is reported. A classical flux-form finite-volume operator instead admits a fully executable conservation verdict (claim C34, supporting artifact).

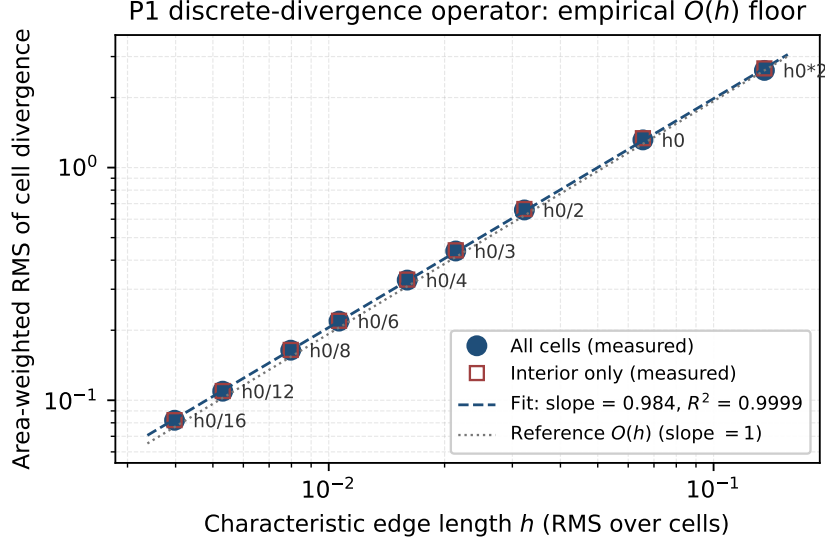


Figure 3: P1 divergence operator-floor calibration.

5.6. Coverage implication: fault-detection robustness across the multi-checkpoint roster

To check whether the primary MGN seeded-fault result of Section 5.2 reproduces beyond one checkpoint and delimit where its detectors are structurally insensitive, the 10-mutant catalogue is replayed against the $K=6$ multi-checkpoint roster across five input-permutation seeds (30 trials per mutant per detector), with two predeclared severity dimensions swept: NS_double_scale $s \in \{1.1, 1.25, 1.5, 2, 4\}$ and PC_zero_vy partial fraction $p \in \{0.25, 0.5, 0.75, 0.85, 0.9, 0.95, 0.99, 1.0\}$ (the canonical mutant is $p = 1.0$). Detection is decided by the same predeclared thresholds as Section 5.2 (node-permutation tolerance 10^{-5} , conservation ratio threshold 1.5, mirror-y relative-change threshold 0.5). Wilson 95% CIs are computed across the (SUT, seed) cells; raw outputs are committed (Table A1). This 10-mutant replay is the primary MGN stress-test denominator, not the whole fault-evidence inventory summarized in Section 4.2.

Table 8 summarizes the coverage geometry that would otherwise be buried in several paragraphs. The table should be read by row: each admitted relation measures one invariant family, and the blind cells are therefore structural, not sampling noise.

Aggregate reading. The 5-of-10 union detection rate from C10 repro-

Table 8: Coverage-geometry matrix.

Detector / relation family	Visible directions	Blind or excluded directions	Scope and boundary
Node permutation	Representation contract passes exactly; also admitted on airfoil	Canonical injected faults preserve node relabeling, so no mutant is detected	Sanity/adapter evidence, not fault-coverage evidence
Continuity / conservation	Boundary-condition faults and gross normalization faults; shared continuity $\rightarrow$ normalization-scale signal on airfoil	Multiplicative output scaling sweep remains undetected (0/6, CI [0.00, 0.39])	Measures conservation-like invariants only; absolute MGN conservation remains deferred
Mirror-y symmetry	Physical-channel and mesh-adjacency faults; partial v_y zeroing detected through $p = 0.99$ (6/6)	Exactly uniform $p = 1.0$ is blind (0/6); mirror-y excluded on non-zero-angle airfoil	Coverage disappears when the relation is inadmissible
Admitted suite	5/10 canonical MGN mutants detected; reproduces across S0–S3, with 4/10 on S4/S5	Invariant-preserving faults remain invisible, including adversarial escapes	Bounded stress-test coverage, not real-world defect-detection rate
Cross-family / cross-program checks	PINN/FNO probes and sibling programs reproduce the invariant-based reading	Trained-mutant MGN/PointMLP evidence and closed-form PINN/FNO probes are not interchangeable	Breadth evidence only; no calibrated coverage law

duces across S0–S3 and nearly across S4/S5. By-class localization also reproduces: continuity is sensitive to boundary/scale faults, whereas mirror-y is sensitive to physical-channel and adjacency faults. The severity sweeps in Table 8 delimit the same point: faults are caught when they perturb a measured invariant and are invisible when they preserve all admitted invariants.

Unified catalogue and adversarial checks. Phase 3 emits a 60-entry catalogue (10 canonical MGN mutants, 2 adversarial MGN mutants, 24 closed-form PINN probes, and 24 closed-form FNO probes). Its descriptive precision/recall summary is node-permutation 1.00/1.00, conservation 1.00/0.81, and mirror-y 0.94/0.55; these rates summarize non-independent checkpoint-by-seed cells, not independent population items. The PINN/FNO probes further show a falsifiable blind region: faults preserving every measured invariant are detected by no relation, even when a transport fault corrupts the PINN field by 0.63 relative L_2 norm. **Adversarial mutants (R4).** The adversarial mutants test whether this blind region is a subspace, not a point: A1 is detected by mirror-y through magnitude shift, with node-permutation 0/6 and conservation 0/6, so it is not evidence that the node-permutation or conservation detectors cover the blind subspace; A2 escapes

every detector.

Cross-SUT and cross-program breadth. On the bounded PhysicsNeMo airfoil task (claim C31), the gate excludes mirror-y because of non-zero angle of attack, and the mirror-only cylinder-flow coverage disappears; the shared continuity $\rightarrow$ normalization-scale signal remains. This supports the qualitative validity-coverage duality, but only at fault-class level on two CFD SUTs. The Minimum-MR-SubSet sibling adds an external breadth audit across seven program types and five central-processing-unit (CPU)-only replays; Appendix Appendix B gives the details.

Independent periodic-advection primary workflow. The **periodic-advection primary workflow** adds a new synthetic SciML surrogate task rather than another cylinder, airfoil, PINN, or FNO extension. Six deterministic NumPy-trained periodic-convolution SUTs are trained for one 2D scalar periodic-advection step on a 32 x 32 periodic grid and evaluated on 10 held-out fields per SUT. The run records relation records, trained checkpoints, source/follow-up output archives, per-SUT metric ledgers, gate decisions, and an aggregate verdict report. Periodic integer translation is admitted and gives **60/60 translation passes** (maximum relative- L_2 violation 0.0 at tolerance 10^{-10}). Global mass/mean conservation is admitted and gives **60/60 mass-conservation passes** (maximum RMS-normalized mean drift 3.47×10^{-17} at tolerance 10^{-10}). The **fixed-velocity mirror candidate is rejected** for 6/6 SUTs and is not executed as an exact MR because mirroring the scalar field without transforming the advection velocity changes the transport problem. This is full gate-to-verdict evidence on an independent primary-scale synthetic PDE task (claim C54), **not production CFD or real-defect evidence**.

5.7. Interpretation and boundaries

The value of the current study is not that every MR finds a new fault beyond rollout accuracy. Retained MRs provide relation-level evidence under explicit transformations; on a violation or deferral, the relation record and verdict rule help distinguish model inconsistency, relation-domain boundary, numerical tolerance issue, and inconclusive evidence.

Table 9 separates the interpretation licensed by the evidence from claims that remain outside scope.

Two misreadings are especially important. First, without the gate, asymmetric-mesh mirror-y could be read as a model fault rather than an out-of-relation-domain stress result; with the gate, it remains a robustness

Table 9: Interpretation and claim-boundary guide.

Reader concern	Evidence response	Licensed claim	Not licensed
Does MR evidence add value beyond accuracy?	Rollout accuracy and mirror-y violation measure different objects; seeded faults include MR-only catches	Relation-indexed checks complement accuracy monitoring	Admitted relation set superiority over accuracy monitors
Is the gate merely post-hoc labeling?	Symmetric-mesh, airfoil, PINN, FNO, and witness reruns apply the same predicate outside the first pilot	The predicate changes or preserves verdicts according to relation conditions	General reliability from one MeshGraphNets family
Does downgrading hide robustness failures?	OOD verdicts remain reported as stress or robustness evidence, but not as valid-domain fault verdicts	Gate controls fault attribution	Using domain invalidity to excuse deployment weakness
Is deferred conservation a pipeline failure?	FNO periodic conservation and PINN reference-relative conservation produce executable verdicts where the floor is calibratable	The gate fails closed when numerical decidability is missing	Absolute MGN mass conservation on the coarse mesh
What overclaims are blocked?	Typed verdicts distinguish failure, OOD stress, rejection, deferral, and inconclusive evidence	Bounded V&V utility with auditable artifacts	Geometry-independent rates, validated localization, or real-world defect detection
When is the method worth using?	A relation must be physically meaningful and numerically decidable for the deployed discretization	Use for relation-indexed OOD or physics checks alongside accuracy/UQ	Replacement for residual, UQ, or accuracy diagnostics
What is the cost profile?	Relation-record authoring and floor analysis are one-time per-relation design costs; checks then replay automatically	Reusable executable checks with visible assumptions	Zero-cost automatic MR discovery or measured person-hour savings

signal but not a valid-domain defect oracle. Second, without the floor check, absolute discrete-divergence could be misread as a conservation pass despite reference-field divergence. The added runs answer whether this is only post-hoc labeling: the same predicate admits, rejects, or defers relations differently on symmetric meshes, airfoil, PINN, and FNO subjects, while preserving the stated reliability and geometry-independent-rate boundaries.

Boundary of claims.

The canonical blocked list remains active: the evidence does not support external-dataset or geometry-independent pass/fail rates, AeroGraphNet/DoMINO results, comparative superiority over baselines, real-world

fault-detection rates, validated localization, runtime, reliability, or model accuracy claims. The supported claim is that an admissibility-gated workflow makes MR identification and execution more auditable for the studied class of SciML SUTs.

Implications for SciML testing.

The scoped evidence shows how SciML testing can move from implicit expert checks to explicit relation records and executable checks that complement residuals, UQ, and accuracy by making transformations and verdict rules inspectable. The workflow targets a relation-indexed applicability map in relation space rather than residual space; calibrated maps across SUTs, trajectories, and domain-violation scores remain future work.

When to use, and at what cost. The gate earns its overhead when a surrogate must be trusted out of distribution and a physically necessary relation can be both stated for the task and made numerically decidable for the deployed discretization. Authoring an admissible MR requires domain knowledge to state the relation, numerical analysis to set or reject the floor, and testing expertise to bind transformations, mappings, metrics, and ledger fields. This paper decomposes those roles and provides reusable records and runners, but it does not measure person-hours, inter-author agreement, or maintenance cost; those are adoption questions for future work.

6. Threats to Validity

We classify threats following Verdecchia et al. [46] and report against the Empirical Standards for software engineering research [47].

Construct validity. The main construct is not “model correctness” but admissibility-gated MR checking. A wrong physical premise, boundary-condition reading, output mapping, discrete operator, or tolerance could therefore change an admit/reject/defer decision. We mitigate this by requiring a relation record, explicit preconditions, typed verdicts, and an operator-floor argument before a relation is treated as executable evidence. The mitigation is still incomplete: the P1 divergence floor is derived for the deployed symmetric-mesh setting and is empirically stable on a second unstructured Delaunay topology but is not proven for arbitrary unstructured meshes, and the reference-relative conservation diagnostic remains a non-regression guard rather than an absolute conservation measurement. The seeded-fault catalogue also measures detector sensitivity to author-implemented mutants, so its rates do not estimate real-world defect prevalence or effectiveness. A

gate verdict can prevent invalid fault attribution, but it must not be read as dismissing deployment robustness concerns outside the relation domain.

Internal validity. SUT setup, checkpoint selection, random seeds, mesh preprocessing, normalization, and runtime nondeterminism may affect verdicts. The study records manifests, ledgers, and raw outputs, and it repeats the primary cylinder-flow checks across $K=6$ checkpoints, three held-out trajectories, same-task variants, PointMLP, and PhysicsNeMo. These repetitions reduce the risk of a single-run artifact, but do not make the cells independent population samples. For airfoil, high one-step rollout error is read as training-state context, not model accuracy evidence.

External validity. The evidence covers incompressible cylinder flow, compressible airfoil flow, bounded PINN/FNO Burgers and heat executions, an independent periodic-advection primary workflow over six synthetic periodic-convolution SUTs, and a read-only/executed cross-program breadth check from the Minimum-MR-SubSet sibling artifacts. These cases make the gate less likely to be a one-off cylinder-flow checklist, but they remain far from a general claim about SciML surrogates. The periodic-advection run adds a complete gate-to-verdict chain on a different PDE/operator setting; it does not convert the evidence into production CFD, real-defect, or reliability rates. Broader neural operators, mesh simulators, production CFD solvers, PINN architectures, datasets, geometries, and training regimes remain outside the evidence. The cross-program breadth check reuses the sibling’s domain evidence and mutant sets, so it supports only a qualitative coverage-geometry reading, not per-program reliability, new MR discovery, superiority, or real-world fault-detection rates.

Baseline and comparator validity. Rollout accuracy, residual/UQ-style diagnostics, generic MR templates, and LLM-assisted candidate generation are used as scope contrasts rather than defeated competitors. The LLM baselines depend on prompt design, model versions, and simulated raters; they show why an admissibility gate is needed, but are not a human-expert study and do not establish superiority over alternative MR-identification methods. The deterministic admissibility gate, not LLM output, is the validity authority.

Conclusion validity. The study combines descriptive counts, Wilson intervals, bootstrap intervals, and exact nonparametric tests where the sampling unit supports them. Many reported cells share checkpoints, trajectories, or generated cases, so are not independent population trials. The primary MGN scope upgrade improves the weakest denominators within the stated

scope, but the correct conclusion is bounded empirical utility for RQ4, not a calibrated reliability or coverage rate.

Reproducibility. Some workflows depend on old runtimes, public benchmark loaders, or non-redistributable checkpoints. The replication package therefore emphasizes scripts, manifests, run logs, claim ledgers, and available outputs. Reproducibility is strongest for artifact inspection and rerunnable CPU workflows, weaker for exact recreation of every trained checkpoint environment. By re-run cost: the operator-floor sweep, relation-record validators, and typed-verdict logic replay on CPU in minutes from committed scripts alone; the cylinder-flow MGN pilot, K=6 roster, PointMLP, and FNO/PINN rosters replay on CPU from committed checkpoints and metric ledgers; PhysicsNeMo MGN and airfoil training require a graphics-processing unit (GPU) and the public DeepMind TFRecords; and the Minimum-MR-SubSet audit and rerun execute read-only from sibling fixtures at the cited commit.

7. Future Work

Future work should generalize the numerical-decidability theory beyond shape-regular P1 triangular settings to degenerate/sliver-heavy meshes, boundary treatments, flux-form operators, discontinuous fields, and learned-output-specific floor models. The present evidence supports a local shape-regular P1 bound, a tighter closed-form and bounded floor for the concrete structured setting, and observed stability on one Delaunay topology; it does not provide an arbitrary-mesh theorem.

Second, the domain-violation axis should be calibrated across MR families. Here, domain distance is a per-relation interpretive coordinate, not a cross-MR score. A calibration protocol would normalize distances within relation families, define in-domain/near-boundary/out-of-relation-domain anchor cases, build relation-family calibration sets, and use independent experts or raters to validate categories, agreement, and residual comparability limits.

Third, the relation-record workflow should be evaluated on production-scale SUTs and datasets: external aerodynamics, production graph simulators, longer rollouts, and multi-physics pipelines.

Fourth, the coverage implication should be tested with independent real-defect corpora and independently authored mutants, asking whether admitted MR sets predict externally collected defect classes.

Finally, relation-record authoring, precondition checking, and ledger generation need automation and user studies of authoring effort, role division, inter-author agreement, and maintenance cost, while validity decisions remain tied to physical, numerical, and software evidence rather than model consensus.

8. Conclusion

This paper asked when a physics-derived MR for SciML surrogate software is sound enough to use as an oracle-free test. The answer is a numerical-decidability admissibility criterion embedded in an admissibility-gated workflow: a candidate relation is first checked for physical basis, domain preconditions, representation mapping, and numerical decidability; admitted relations are recorded as relation records and executable checks; and outcomes are interpreted with typed verdicts rather than a single pass/fail label.

The empirical evidence is bounded but artifact-backed. On cylinder-flow MeshGraphNets, the workflow separates a representation pass, an out-of-relation-domain mirror probe, a deferred absolute conservation relation, and a reference-relative diagnostic. The operator-floor analysis shows why the conservation verdict is not numerically decidable as an absolute test in that setting, while alternative settings admit executable conservation checks. On airfoil, PINN, FNO, and cross-program subjects, the gate changes or preserves verdicts according to the relation’s physical and numerical conditions. Seeded-fault stress tests further show that the admitted MR set helps explain detector blind spots in the studied settings, without implying general defect-detection rates or excusing OOD robustness failures.

The contribution is therefore methodological and evidentially bounded: physically meaningful SciML metamorphic tests require explicit validity conditions, a numerically decidable verdict, executable checks, raw evidence records, and relation-level verdicts. This turns candidate MRs from informal expert intuitions into auditable tests with boundaries visible to reviewers and practitioners, while leaving arbitrary-mesh theory and general defect-rate evaluation as open work.

Preprint

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CRedit authorship contribution statement

Meng Li: Conceptualization, Methodology, Software, Writing – original draft. **Xiaohua Yang:** Supervision, Formal analysis, Writing – review & editing. **Jie Liu:** Investigation, Validation. **Shiyu Yan:** Data curation, Visualization.

Declaration of competing interest

The authors declare no conflict of interest.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used generative AI assistants for code scaffolding, prose copy-editing, and reference cross-checking. After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No generative AI was used to fabricate, alter, or invent experimental data, results, or citations; every numerical claim is backed by a tracked artifact under `research_assets/runs/` and a claim-ledger entry.

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Data availability

The replication package is archived on Zenodo at <https://doi.org/10.5281/zenodo.20702952>, with source at <https://github.com/meng004/Domain-Validity-Gated-MR-for-SciML>. It contains the manuscript source, relation records, validators, claim ledger, run manifests, metric ledgers, and committed derived outputs under `research_assets/runs/`. Minimum-MR-SubSet audit/rerun artifacts cite commit `9ef862ec37335b4834d0a1fb38b4b613af702f34`. DeepMind cylinder-flow and airfoil TFRecords are public

benchmark inputs staged by the workflow runners rather than redistributed in the manuscript package. GPU-dependent reruns and credential requirements are documented in `REPRODUCIBILITY.md`; absent credentials cause fail-closed precondition checks rather than silent substitution.

Appendix A. Claim-to-evidence map

Table A1: Claim-to-evidence map.

Claim	Status	Evidence	Boundary
Admissibility gate	Method claim	Gate definition and manuscript method section	Establishes an auditable screening rule; physical validity still depends on each relation's stated preconditions.
Relation-record executable checks	Workflow claim	Relation records and validators	Some records remain protocol evidence pending matched SUT evidence.
Baseline admissibility contrast	Observed	Expert-MR, generic-MR, LLM-MR baselines and claim ledger	Quantifies an admissibility gap without ranking methods as competitors.
Three-arm complementarity / gate value	Observed	PointMLP three-arm report and detection-vs-accuracy metric ledger	admitted relation set 13/20 vs accuracy monitor 6/20 (2×2 : 9/2/4/5, Wilson CIs); gate-admitted template 0% vs gate-rejected templates 100% baseline false positive; a measured gate value and complementarity, never a superiority ranking; consolidated across three converged SUTs (cylinder MGN 5/10 vs 3/10, airfoil 1 vs 2 disjoint, airfoil gate-rejected templates 4/4 firing on the correct SUT; C51).
Minimum-MR-SubSet external-scope audit	Observed secondary audit	External-scope audit JSON	external witness evidence from 70 real rows and three SciML/PDE true-fault-class rows with PASS_WITNESS status; does not add new primary SUT executions to this paper.
Minimum-MR-SubSet primary rerun	Observed external runtime rerun	Witness rerun report	PASS_WITNESS status, kstar = 6, four active true fault classes, max signature rank 2, collapse false; real held-out cylinder-flow MGN runtime, not a second architecture or dataset.
Minimum-MR-SubSet PINN primary reruns	Observed second-SUT/PDE reruns	PINN witness rerun reports	trained Burgers2D and Diffusion2D PINN witnesses, PASS_WITNESS status, kstar = 1 each, five active true fault classes each; one-seed witnesses, no cross-SUT rate.
Node-permutation sanity check	Observed pilot	Node-permutation metric ledger	One representation-contract path on one pilot case.
Conservation diagnostic	Diagnostic; absolute relation deferred	Conservation metric ledger and report	Reference-relative guard only; absolute conservation remains deferred.
Mirror-y OOD stress (PC6-mirror-y-ood-stress)	Observed bounded pilot	Mirror-y rate metric ledger and claim ledger	Failed on 10 of 10 recorded eval frames; not a reliability, accuracy, baseline, multi-SUT, exact-symmetry, or geometry-independent claim.

Claim	Status	Evidence	Boundary
Primary empirical scope upgrade	Observed	Primary-scope report and per-trajectory/checkpoint manifests	K=6 x 3 trajectories x 10 mirror-y OOD-stress grid fails 180/180; K=6 x 3 trajectories x 9 conservation-transition grid passes 162/162; K=6 x 3 exact-symmetric-mesh input grid fails 18/18; not a single-source-trajectory estimate and not a cross-SUT rate.
LLM role boundary	Process boundary	Method and ethics sections	LLMs organize candidates; the admissibility gate decides validity.
Rollout-accuracy diagnostic	Observed	Rollout-accuracy metric ledger	Same-SUT one-step median relative L_2 norm 0.0216; mirror-y is $\sim 34\times$ larger.
Exact mirror-y symmetric mesh	Observed	Symmetric-mesh metric ledger	Exact relation admissible and fails (relative L_2 norm 1.10) on a synthetic OOD mesh; binary equivariance evidence.
PC10-seeded-fault-detection	Observed	Seeded-fault metric ledger	Detector stress test: MRs catch 5/10 injected mutants with class-specific response patterns.
Multicheckpoint replication	Observed	E1 aggregate and per-SUT manifests in multicheckpoint/	K=6 checkpoints of one MeshGraphNets family; within-family replication.
Operator-floor resolution	Observed	Operator-floor sweep report	Log-log slope 0.984, 95% CI [0.975, 0.992] (interior 0.989), $R^2 = 0.9999$ over nine resolutions; reproduced on a second unstructured Delaunay topology (slope 0.983, floor ratio 1.06 median / 1.33 max), calibrating the admissibility predicate across two mesh topologies.
Shape-regular P1 operator-floor soundness	Supported theory	Phase B proof artifact and operator-floor reports	Local bound for P1 constant-per-cell divergence on shape-regular triangular meshes with C^2 divergence-free reference fields; supports numerical-decidability soundness for this operator class, not arbitrary degenerate meshes, non-P1 operators, or discontinuous fields.

Claim	Status	Evidence	Boundary
Fault-detection robustness	Observed	E3 robustness report; Phase-3 unified catalogue	30-trial Wilson CIs per MGN mutant plus a 60-entry unified fault catalogue (10 canonical MGN + 2 adversarial MGN + 24 closed-form PINN output-level probes + 24 closed-form FNO output-level probes), by-detector precision/recall with Wilson CIs, and Wilcoxon/Cliff effect-size tests; with realistic non-tailored faults the by-class structure emerges as a two-by-two partition (changes integral $\times$ breaks symmetry) on FNO and PINN, and a falsifiable coverage-completeness holds — every invariant-preserving fault is invisible to the admitted relation set with no falsifying case, blind even at 0.63 relative L_2 norm on PINN; PINN/FNO probes are closed-form output-level probes.
S4/S5 variant primary workflow	Observed	Variant report and raw ledgers	Same-domain wider/deeper MGN variants: 2/2 node-permutation passes, 60/60 mirror OOD failures, 54/54 conservation-diagnostic passes, 6/6 exact-symmetric failures; not PhysicsNeMo/EchoWave or cross-dataset reliability evidence.
PointMLP cylinder primary workflow	Observed	PointMLP report, checkpoint, and raw ledgers	Newly trained non-MGN row-wise coordinate network: 9/9 node-permutation passes, 10/10 mirror OOD failures, 9/9 conservation-diagnostic passes, 3/3 exact-symmetric failures, median rollout relative L_2 norm 0.0298; not PhysicsNeMo/EchoWave or production CFD evidence.
PhysicsNeMo MGN Object-A smoke workflow	Observed smoke-subset production-framework execution	PhysicsNeMo smoke report, newly trained checkpoint, raw outputs, and metric ledgers	NVIDIA PhysicsNeMo MeshGraphNet executes on a first-record official DeepMind cylinder_flow smoke subset: node permutation passes (relative L_2 norm 0.0), mirror-y is OOD stress, and conservation is reference-relative diagnostic only; not full production-scale PhysicsNeMo, AeroGraphNet, or DoMINO evidence.
FNO primary workflow upgrade	Observed	FNO workflow report, per-SUT ledgers, and raw source/follow-up outputs	Six trained FNO-2D checkpoints over Burgers/heat: 24/24 translation passes, 24/24 conservation failures under a periodic discrete-conservation MR, and 6/6 Dirichlet-boundary translation rejections; full execution evidence beyond admissibility.

Claim	Status	Evidence	Boundary
Periodic-advection primary workflow	Observed	Periodic-advection workflow report, relation records, checkpoints, per-SUT ledgers, and raw source/follow-up outputs	Six NumPy-trained periodic-convolution SUTs on a synthetic 2D scalar periodic-advection task: 60/60 translation passes, 60/60 mass-conservation passes, and 6/6 fixed-velocity mirror rejections; full execution evidence on an independent synthetic PDE task, not production CFD or real-defect evidence.

Appendix B. Cross-program breadth

Cross-program generalization. Reusing committed Minimum-MR-SubSet kill matrices, the coverage-geometry reading is checked across seven program types in three families: neural surrogates, classical numerical solvers, and an OpenMC subject run. Fault coverage follows each program’s admitted MR set; structural blind spots remain in six of seven programs; and class-to-class mappings are program-specific. This supports the falsification role of the admitted-set view. It is a read-only check from committed evidence, not an end-to-end execution of this pipeline on each program (claim C39).

End-to-end cross-program execution. The same gate and typed verdict are also executed on five CPU-only sibling SUTs: heat, wave, point-kinetics, Burgers, and the sibling’s OpenMC single-energy-group subject run. Across the five, the gate admits 38 candidate relations and rejects 7 at output-mapping conditions; admitted relations hold on correct programs, while mutant runs populate the SUT-inconsistency region. Detection tracks the admitted set rather than a uniform rate (full-set mutation score 0.75 on heat, 0.20 on the other three solvers, and 0.56 on OpenMC). The MRs and mutants are the sibling’s; only the gate and typed verdict are newly executed here, so this asserts no per-program reliability, no superiority, no new MR contribution, and no continuous-energy or whole-core reactor-physics result (claim C40).

Appendix C. Secondary baselines, external-scope audit, and cross-family executions

Appendix C.1. RQ4 secondary baselines and external-scope audit

This subsection reports the Secondary baseline and external-scope audit; LLM baselines are secondary exploratory scope contrasts, not competitors.

Expert-LLMs proposed 24 unique candidates, admissibility voting retained 4 with no novel family, generic templates admitted 3/13, and one-shot rater agreement was Fleiss kappa 0.077. These numbers concern candidate screening, not the deterministic operator-floor admissibility condition. On the converged PointMLP SUT, the admitted relation set and rollout-accuracy monitor are complementary over a 20-fault catalogue (relation set 13/20, monitor 6/20; MR-only 9, accuracy-only 2, both 4, neither 5). The single gate-admitted generic template has 0% baseline false positives on the correct SUT, whereas all six gate-rejected templates fire on it (100%). This is a measured gate value with descriptive Wilson CIs and no superiority test. The Minimum-MR-SubSet external witness evidence records 70 real witness instances, three SciML/PDE rows with PASS_WITNESS status, and one held-out cylinder-flow MGN rerun; it does not add new primary SUT executions to this paper.

Appendix C.2. RQ4 Cross-family PINN extension and FNO upgrade

The cross-family executions have a supporting falsification role. They test whether the same four admissibility conditions produce typed decisions on pointwise PINNs and spectral FNOs, and whether conservation MRs execute wherever the floor is calibratable: FNO periodic conservation fails 24/24 against a case-level floor and PINN reference-relative conservation passes 30/30, while the MGN open-boundary absolute variant is refused. The cylinder-flow deferral is therefore gate discrimination rather than missing verdict evidence.

The K=15 PINN roster contains fifteen Burgers and fifteen heat seeds. **MR-A** is *vacuous by construction* for pointwise MLPs. The two non-trivial MRs carry the evidence: **MR-B** passes on 13/15 Burgers seeds (mean 0.712, CI [0.583, 0.875]) and is mixed on heat with 7/15 passing (mean 1.495, CI [1.142, 1.897]); **MR-C** passes on all 15/15 of both PDEs, with Burgers mean 1.007 (CI [0.997, 1.017]) and heat mean 1.006 (CI [0.988, 1.022]). This is a two-PDE PINN seed roster rather than a PINN-vs-MGN benchmark or a generality claim across PINN architectures, PDEs, or training regimes.

The **FNO primary workflow upgrade** uses six trained FNO-2D checkpoints over Burgers and heat. Periodic integer translation is admitted and produces **24/24 translation passes** (maximum relative- L_2 violation below 10^{-5}). The **periodic discrete-conservation MR** is admitted-with-reference-floor, and trained FNO outputs exceed that calibrated floor on

24/24 conservation failures. The Dirichlet translation candidate is rejected for 6/6 SUTs because it changes the boundary-value problem and is not executed as an exact MR. This is **not only admissibility evidence**: it is full gate-to-verdict FNO evidence with raw source/follow-up outputs and per-case ledgers, while remaining outside cylinder-flow evidence, performance benchmarking, reliability, and broad neural-operator generalization. The separate FNO admissibility roster is admitted on 15/15 per PDE (Wilson 95% CI [0.80, 1.00]), with Burgers mean violation 4.5×10^{-8} and heat 1.4×10^{-7} .

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